

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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Sequential strain and doping scans are standard in covalent molecular networks, but total-energy effects need not add. In quasi-hexagonal C₆₀ graphullerene we quantify the omitted cross term with a synergy order parameter $\mathcal{S}$ (meV/atom). PBE+DFT-D3 on tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$) gives opposite strain coefficients for N (-303 meV/%) and B ($+57$ meV/%) and $|\mathcal{S}|$ up to ~ 32 meV/atom at $+3\%$ strain, largest for P despite small $|\alpha|$. The pattern warns that sequential high-throughput screens can mis-rank stability when local dopant distortion couples nonlinearly to external strain.

INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only calculations [1, 2]. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications, yet charge transport in molecular networks is often polaronic [3, 4]. The quasi-hexagonal phase (qHP C₆₀, graphullerene) [4] covalently links C₆₀ units, shortening interfullerene distances by ~ 0.9 Å and raising mobilities by nearly two orders of magnitude over vdW C₆₀ [3]. Capobianco *et al.* [3] showed that the gain tracks enhanced nearest-neighbor coupling J ($\sim 75\text{--}81$ meV vs. 33 meV) within a hopping model, with similar polaron binding ($\lambda^- \sim 0.1$ eV) in both phases.

Beyond pristine monolayers, strain and substitution independently tune qHP C₆₀ band edges [2, 5, 6]. Non-additive strain–doping coupling has been reported in other two-dimensional networks [1]; a quantitative energy cross term $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells under a common PBE+D3 protocol. If formation energies or strain penalties are estimated from *separate* doping-only and strain-only scans, the omitted cross term can reach tens of meV/atom in coupled cells (this work)—large enough to mis-rank dopant stability or strain set-points in screening workflows.

We combine Capobianco-style observables (gaps, IPR, J) with systematic strain–doping energy scans on shared tetramer and size-scaling grids. The present study establishes the *energy* coupling metric $\mathcal{S}$ and band-edge trends; Marcus λ and rate estimates are treated in Supplemental Material and are not required for the non-additivity conclusions here.

COMPUTATIONAL METHODS

Electronic-structure setup

All total-energy and electronic-structure calculations use CP2K [7] with PBE [8] and Grimme DFT-D3 (Becke–Johnson damping) [9], as documented in Supplemental Material. This level is deliberately chosen for a uniform, size-converged scan of (ϵ, δ) ; it differs from the Koopmans-compliant PBE(α_K)+rVV10 framework of Ref. 3, which targets absolute polaron binding and mobilities in pristine supercells.

Rigid-strain approximation. Unless stated otherwise, biaxial strain is imposed by scaling the in-plane lattice vectors of the *pristine* equilibrium cell at *fixed fractional atomic coordinates* (no atomic relaxation under load), not by re-scaling each dopant-relaxed equilibrium lattice. Reported α therefore mixes dopant-induced changes in strain response with any equilibrium lattice offset relative to pristine. This protocol upper-bounds the strain energy and is applied consistently across the comparative grids below. In related 2D carbon networks, allowing ionic relaxation under load typically reduces strain-energy derivatives relative to rigid cells [2]; we therefore interpret reported α and $\mathcal{S}$ as *upper bounds* on the rigid-lattice coupling, with relaxed-ion values expected to be smaller in magnitude but not necessarily vanishing (ionic-relaxation benchmark in Supplemental Material Table S3, pending GEO_OPT).

Plane-wave cutoff is 300 Ry for cells with $n \leq 5$ fullerene units and 280 Ry for $n \geq 6$ (memory limit for 6×8 cells on our hardware). Because $\mathcal{S}(n)$ compares energies *within* each n at identical cutoff, trends at fixed n are internally consistent. Cross- n comparisons—including the N $\mathcal{S}$ sign change between $n = 6$ and $n = 8$ —coincide with the 300→280 Ry switch and should be interpreted cautiously until a matched-cutoff check at $n = 6$ is available (Supplemental Material Table S2).

SCF convergence criterion 10^{-6} Eh with OT minimization; basis DZVP-MOLOPT-SR-GTH with matching GTH pseudopotentials.

Substitutional doping protocol

Tetramer models ($4 \times \text{C}_{60}$, 240 C sites) use substitutional B, N, or P at 5% of carbon sites (~ 12 replacements per cell), chosen by a reproducible pseudo-random protocol (fixed seed 42) over all C coordinates—not a symmetry-averaged or thermodynamic minimum. Periodic size-scaling cells place **one** substitutional dopant per C_{60} unit (fixed count, so nominal concentration $\sim 1/60$ per molecule decreases as n increases). The two protocols differ in *nominal dopant concentration* (5% vs. $\sim 1.7\%$ at $n = 4$); tetramer α , gaps, and IPR/ J are therefore trend indicators, not directly interchangeable with $n=4$ periodic energies without a matched-concentration benchmark. We do not re-optimize dopant sites; all reported α and $\mathcal{S}$ refer to these fixed, reproducible configurations and should be read as site-specific trends pending tests on alternate placements or thermodynamic minima.

Computational grids

In total, $\gtrsim 90$ converged CP2K tasks support the main-text claims (Table S2); the principal neutral-energy grids are:

Mechanical strain response (24 calculations). Pristine and B/N/P tetramers at $\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$ supply total energies for $\alpha = dE/d\epsilon$ and substitution energies E_f (Table S1).

Electronic structure (12 calculations). The same tetramer compositions at $\epsilon \in \{-5, 0, +5\}\%$ yield π -projected DOS, HOMO–LUMO gaps, and band-edge energies.

Polaron localization (2×2 factorial, four calculations). IPR and nearest-neighbor coupling J are evaluated on a 2×2 grid: pristine vs. B doping crossed with $\epsilon \in \{0, +3\}\%$ (converged neutral single-points; Supplemental Material Fig. S6).

Charged polarons (Supplemental Material). Tetramer GEO.OPT at $q = \pm 1$ and vertical single-points at neutral geometry target Marcus $\lambda^\pm$ [3]; they are not used for $\mathcal{S}$ or α in the main text.

Non-additive size scaling (40 calculations). Periodic cells at $\epsilon \in \{0, +3\}\%$ for each dopant and $n = 1\text{--}8$ supply $\mathcal{S}(n)$; dopant count is one site per C_{60} unit (Table S2).

Reference geometries (Supplemental Material). Neutral tetramer relaxations and a pristine single-point on the relaxed cell define charged-state input geometries.

Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table S1). Positive α means tension *raises*

total energy; negative α means tension *lowers* it under the rigid-strain protocol. Substitution energy $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is reported relative to pristine tetramers at the same strain (no chemical-potential reference). In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_f(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows, the magnitude of this error is best gauged against the substitution-energy scale: at $n = 4$ and $+3\%$ strain, P has $|\mathcal{S}| \approx 24$ meV/atom while the per-atom substitution scale is $|E_f| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_f| \approx 16$ eV per P site, $n_{\text{dop}}=12$, Table S1)—i.e. $|\mathcal{S}|/|E_f|_{\text{per atom}} \sim 3\%$, large enough to reorder stability margins in sequential doping-then-strain screens even though $|\mathcal{S}|$ exceeds the rigid per-atom strain increment ($\alpha \times 3\%/N_{\text{atoms}} \lesssim 0.3$ meV/atom for P).

Non-additive coupling metric. We quantify strain–doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times \text{C}_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

To first order in strain, $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon$ when $\alpha_\delta = dE/d\epsilon$ is evaluated for the doped cell and α_0 for the pristine reference at the same n (rigid-strain protocol); this links the mechanical coefficient α to the cross term at small ϵ .

Periodic-cell $\mathcal{S}$ values are obtained from archived total energies via post-processing of converged CP2K outputs (Supplemental Material Table S2). We report $\mathcal{S}$ for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

Localization and coupling (Supplemental Material Fig. S6). Following Capobianco *et al.* [3], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR follows Supporting Information Eq. (S8), $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$, on a Mulliken-projected orbital grid; J is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75\text{--}81$ meV reported for pristine qHP networks [3], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding (Supplemental Material). Marcus reorganization energies follow the vertical-adiabatic decomposition [3]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Supplemental Material Fig. S5.

RESULTS

Results focus on total-energy non-additivity under coupled (ϵ, δ) tuning: band-edge trends provide context, mechanical α coefficients set the linear strain scale, and $\mathcal{S}$ quantifies the cross term omitted in sequential scans (Supplemental Material for IPR/ J and Marcus λ).

Electronic structure under strain and doping

Figure 1(a) reports HOMO–LUMO gaps from tetramer single-points at $\epsilon \in \{-5, 0, +5\}\%$. B introduces hole states near the valence manifold (acceptor character), compressing the π gap toward zero at $\epsilon = 0$, whereas N donates electrons into the conduction manifold (donor character) and P (larger covalent radius) perturbs the network more heterogeneously—hence the dopant-specific gap slopes in panel (a). Figure 1(b) shows π -DOS for B @ $\epsilon = 0$.

Polaron localization and electronic coupling

Supplemental Material Fig. S6 reports a 2×2 factorial of inverse participation ratio and nearest-neighbor coupling J (pristine/B $\times \epsilon=0, +3\%$) in the Capobianco observables vocabulary [3]. Transport-level non-additivity in J is *not* numerically equivalent to total-energy $\mathcal{S}$ and is not used to support the energy-cross-term conclusions below; the main-text panel (d) inset is illustrative only.

Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Substitution opens a wide spread: N ($\alpha = -303.2$ meV/%, tension stabilizes), B (+56.9 meV/%, tension destabilizes), P (−19.1 meV/%) (Table S1). Figure 1(c) highlights the opposite α for N vs. B; ranking dopants by $\epsilon = 0$ stability alone is misleading.

Non-additive strain–doping coupling

Figure 1(d) tracks $\mathcal{S}(n)$ with one dopant per C_{60} . At $n = 4$ and +3% strain, $\mathcal{S} = -6.2$ meV/atom

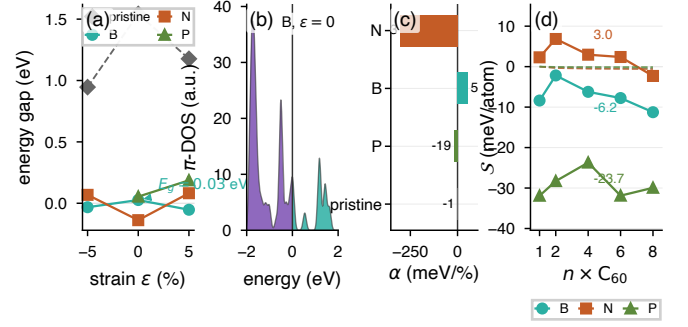


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} (PBE+D3). (a) HOMO–LUMO gap vs. strain. (b) π -DOS for B @ $\epsilon=0$ (Electron SI palette). (c) Mechanical α . (d) $\mathcal{S}(n)$ at +3%; inset: schematic IPR/ J grid (details in Supplemental Material Fig. S6).

(B), +3.0 meV/atom (N), and −23.7 meV/atom (P); P shows the largest magnitude. Fifteen coupled points span $n \in \{1, 2, 4, 6, 8\}$ and B/N/P at +3%. For N, $\mathcal{S}$ changes sign between $n = 6$ (+2.4 meV/atom) and $n = 8$ (−2.3 meV/atom). Because $n = 6$ –8 use 280 Ry whereas $n \leq 5$ use 300 Ry, this reversal may partly reflect basis-set change rather than intrinsic size physics; we treat it as provisional pending a matched-cutoff $n = 6$ benchmark.

DISCUSSION

Relation to Ref. 3. Capobianco *et al.* [3] compared *pristine* vdW and qHP C_{60} and identified enhanced J —not reduced λ —as the dominant mobility lever, with polaron hopping retained in both phases. Our study transfers that observables vocabulary (gap edges, IPR, J) to *doped* tetramers under strain and adds an energy cross term $\mathcal{S}$ absent from sequential doping-only or strain-only scans [2, 5]. Where Khan and Leuenberger [5] classify impurity-engineered *half-semiconductor* gaps at fixed geometry, our $\mathcal{S}$ analysis tests total-energy additivity under identical biaxial strain—complementary metrics for impurity–strain co-design [1]. The factorial J pattern (B raises J at $\epsilon=0$ but +3% strain partially reverses it) is qualitatively aligned with Capobianco’s distance-coupling picture: B acceptor states enhance interfullerene hybridization, making J more distance-sensitive, so tensile strain that increases spacing suppresses J far more under B doping than in pristine cells. Transport and total-energy non-additivity are related in origin but not one-to-one; $\mathcal{S}$ should not be used to infer hopping-rate changes without independent J/λ data.

Mechanistic synthesis. (i) **Opposite α for N vs. B.** N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α)—consistent with

donor vs. acceptor π -counting and the PDOS shifts in Fig. 1(a,b). **(ii) Large $|\mathcal{S}|$ for P despite small $|\alpha|$.** P’s covalent radius (~ 107 pm vs. ~ 77 pm for C) and longer P–C bonds impose intrinsic cage distortion; rigid external strain couples to this built-in local stress, yielding a cross term that is invisible in separate scans but large when strain and substitutional doping are applied jointly—consistent with P’s intermediate E_f yet largest $|\mathcal{S}|$ at +3%. **(iii) N $\mathcal{S}$ sign change with n .** With one dopant per C_{60} , electrostatic and strain fields are dilute at large n ; the sign flip between $n = 6$ and $n = 8$ may reflect finite-size electrostatics but is confounded with the 280 Ry cutoff introduced at $n \geq 6$ —interpretation is deferred until a constant-cutoff control.

(iv) Linear $\mathcal{S}$ – α check. Using tetramer α (Table S1) in $\mathcal{S} \approx (\alpha_\delta - \alpha_0)\epsilon$ at $\epsilon = +3\%$ and dividing by 240 sites gives $\sim +0.7$, -3.8 , and -0.2 meV/atom for B/N/P—far smaller than the periodic $n=4$ values (-6.2 , $+3.0$, -23.7 meV/atom). The gap shows that tetramer-scale linear mechanics and 5% doping do not quantitatively predict periodic $\mathcal{S}$; for P, nonlinear coupling dominates beyond first-order $(\alpha_\delta - \alpha_0)\epsilon$.

(v) Significance versus DFT uncertainty. At $n=4$ and +3% strain, $|\mathcal{S}| \sim 24$ meV/atom is $\sim 3\%$ of the per-atom substitution-energy scale (Table S1) but still $\gg$ typical rigid per-atom strain increments from α . PBE formation energies for molecular solids can carry $\mathcal{O}(10^2)$ meV/atom uncertainties; we therefore treat $\mathcal{S}$ as a *coupled-scan correction* to sequential workflows rather than an absolute formation-energy prediction, and flag cases where $|\mathcal{S}|$ exceeds the rigid strain increment as the primary screening hazard.

Design implications. At +3% strain, P carries the largest additive-approximation error in $\mathcal{S}$ ($|\mathcal{S}| \sim 24$ – 32 meV/atom); co-design should treat P under tension as a *coupled* stability problem, not $|E_f|$ at $\epsilon = 0$ alone. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. Gap closing [Fig. 1(a)] does not track $|\mathcal{S}|$; we do not compute $\mathcal{S}_{\text{gap}}$, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone. N $\mathcal{S}$ sign reversal by $n = 8$ cautions against tetramer-only extrapolation.

Limitations and outlook. Rigid strain (upper-

bound on $\mathcal{S}$; ionic-relaxation check Table S3 pending); fixed-seed substitutional sites without thermodynamic ranking; mixed 280/300 Ry cutoffs across n (N $\mathcal{S}$ sign change at $n = 6$ – 8); tetramer vs. scaling concentration mismatch; PBE+D3 without Koopmans/rVV10; gap non-additivity ($\mathcal{S}_{\text{gap}}$) not quantified; Marcus λ and vertical transitions from charged geometry optimization still pending (Supplemental Material).

CONCLUSIONS

Using qHP C_{60} as a model covalent molecular network, we quantified non-additive strain–doping coupling in total energy. Opposite α for N vs. B and finite $\mathcal{S}$ up to ~ 32 meV/atom show that joint (ϵ, δ) tuning is not equivalent to sequential strain and doping scans; for P under tension, the cross term is a few percent of the per-atom substitution-energy scale yet can significantly alter stability rankings in sequential screening workflows. These energy-landscape metrics complement—but do not replace—Marcus λ from charged-state vertical transitions needed to connect $\mathcal{S}$ to polaron hopping rates.

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- [1] S. Chen, H. Liu, and K. Wang, *Advanced Materials* **36**, 2301234 (2024).
- [2] H. Li, M. A. Shakoori, and R. Wang, *Journal of Applied Physics* **136**, 015101 (2024).
- [3] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, *Nano Letters* **24**, 7891 (2024).
- [4] Y. Yang, W. Zhang, H. Liu, and K. Li, *Applied Surface Science* **536**, 147815 (2021).
- [5] M. A. Khan and M. N. Leuenberger, *Physical Review Materials* **9**, 034001 (2025).
- [6] B. Peng and M. Pizzochero, *Chemical Communications* **61**, 1234 (2025).
- [7] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, *Computer Physics Communications* **167**, 103 (2005).
- [8] J. P. Perdew, K. Burke, and M. Ernzerhof, *Physical Review Letters* **77**, 3865 (1996).
- [9] S. Grimme, S. Ehrlich, and L. Goerigk, *Journal of Computational Chemistry* **32**, 1456 (2011).